Python for Data Analysis: class error glossary

Start list, one row per error card. Students add the errors they meet.

You see	It means	Fix	Session
SyntaxError: invalid character	A curly quotation mark was pasted in. Python only accepts straight quotation marks.	Delete the quotation marks and type them again from the keyboard.	DA 01
FileNotFoundError: camp_shop.csv	The notebook cannot find the file because it sits in a different folder.	Move the notebook into the folder that holds camp_shop.csv, or ask the instructor to help.	DA 01
NameError: name 'price' is not defined	Python has not run a line that creates that name in this session.	Run the cell that creates the name first, and check the spelling and capital letters.	DA 02
TypeError: can't multiply sequence by non-int of type 'float'	You tried to multiply text by a decimal.	Convert the text with float() or remove the quotation marks.	DA 02
ZeroDivisionError: division by zero	You divided by zero, which has no answer.	Check the number you divide by. Change it or guard against zero in a later session.	DA 03
NameError: name 'units' is not defined	The name was never created in this session or is spelled differently.	Run the cell that creates it and compare the spelling letter by letter.	DA 03
IndentationError: expected an indented block	The line under if or else has no spaces in front of it.	Add four spaces to every line that belongs to the if.	DA 04
SyntaxError: invalid syntax at if budget = 5000	A single = was used where a question needs ==.	Use == or >= to ask a question. Use = only to store a value.	DA 04
IndexError: list index out of range	You asked for a position beyond the end of the list.	A list of six values has positions 0 to 5. Check len().	DA 05
KeyError: 'Price'	The key is not in the dictionary, often because of a capital letter.	Check the exact spelling and capitals of the key.	DA 05
AssertionError	The statement after assert was False, so the function gave a different answer from the one you expected.	Print the function result for that input and compare it with your hand calculation.	DA 06
The function shows None	The function has no return line, so it gives nothing back.	Add return in front of the value you want back.	DA 06
FileNotFoundError: [Errno 2] No such file: 'sunrise_sales.csv'	Python cannot find the file where it looks.	Put the notebook and the CSV in the same folder, and check the spelling of the file name.	DA 07
ModuleNotFoundError: No module named 'pandas'	This laptop is not using the prepared environment.	Ask the instructor. Do not install random packages.	DA 07
KeyError: 'unit'	The column name is spelled differently from the header.	Print list(shop.columns) and copy the exact name.	DA 08

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The filter returns an empty table	The comparison matched no rows, often because of a typing slip in a text value.	Check the spelling and capital letters of the value you compare with.	DA 08
KeyError: 'revenue_ugx'	You grouped a column that has not been created yet in this session.	Run the cell that creates the revenue column before the cell that groups it.	DA 09
The totals do not add to the overall total	Rows were lost or counted twice before grouping.	Compare len() before and after each step to find where rows changed.	DA 09
SettingWithCopyWarning	You changed a slice of another table, so pandas is not sure which table you meant.	Make a real copy with .copy() before you change a table.	DA 10
Row counts differ between steps	A step removed or added rows without you noticing.	Print len() after every step and write the numbers in the log.	DA 10
The chart appears empty	plt.show() ran before the data was added, or the data was empty.	Run the cell that creates the table first and print its length.	DA 11
The labels on the dates overlap	There are too many labels to fit side by side.	Rotate them with plt.xticks(rotation=90) or plot fewer dates.	DA 11
The numbers differ after a restart	A cell relies on a value that was created in a cell that no longer runs first.	Restart the kernel and run all cells from the top. Fix the first cell that fails.	DA 12
The total does not match 6375900	A filter, copy or merge changed the number of rows.	Print len() after each step and find where the count changes.	DA 12
TypeError: 'tuple' object is not callable	shape is a value you read. It takes no brackets, and brackets try to call it like a function.	Remove the brackets after shape.	DA 13
The mean from NumPy and pandas differ in the last digits	Decimals are stored with tiny rounding differences.	Compare with np.allclose, or round both to the same number of places.	DA 13
NameError: name 'sns' is not defined	seaborn was not imported in this session.	Run import seaborn as sns in the first cell, then run the chart cell.	DA 14
ValueError: Could not interpret value for x	The column name in x= or y= is not in the DataFrame.	Print list(daily.columns) and copy the exact name.	DA 14
KeyError: "hue"	The hue column name is not in the table.	Print list(daily.columns) and use the exact name.	DA 15
The heatmap shows all one colour	vmin and vmax were not set, or only one column was used.	Pass the four numeric columns and set vmin=-1 and vmax=1.	DA 15
ValueError: time data does not match format	A date is written in a form the format string does not describe.	Use errors="coerce" for one format and fill the gaps with the other format.	DA 16
The clean table has 1538 rows, not 1513 or 1473	The duplicates were not removed before the other steps.	Call drop_duplicates() first and print len() after every step.	DA 16

You see	It means	Fix	Session
pandas.errors.MergeError: Merge keys are not unique	validate found a key that appears more than once in the table you look up.	Find the repeated key with duplicated, fix the table, and merge again.	DA 17
The merged table has more rows than the sales table	A key appears twice in the second table, so rows were copied.	Compare len() before and after every merge and check the key is unique.	DA 17
The category totals add to 6375900, not 6250500	The clean table was not used, or the excluded rows were put back by a merge.	Total from the cleaned table and compare with len() at each step.	DA 18
MergeError or a larger row count after the merge	The key is repeated in the products table.	Check is_unique on the lookup table before you merge.	DA 18
The plan promises profit	The data has no costs, so profit cannot be worked out.	Change the measure to revenue and say so in the limitations.	DA 19
A question needs a column that does not exist	The question asks for something the data never recorded.	Narrow the question to the columns you have, or choose another dataset.	DA 19
A calculation changes when the notebook is restarted	It depends on a value from a cell that no longer runs first.	Restart and run all. Move the first failing cell's dependency above it.	DA 20
The partner gets a different answer	A step uses a file path or setting that only works on your laptop.	Use paths that sit beside the notebook and state them in the first cell.	DA 20
The saved picture cuts off the labels	The figure was saved without room for the labels.	Add bbox_inches="tight" to plt.savefig.	DA 21
The picture is saved blank	plt.show() ran before plt.savefig, which clears the figure.	Call plt.savefig before plt.show.	DA 21
A claim in the report has no number behind it	The sentence was written from memory and not from an output.	Find the output cell that supports it, or remove the claim.	DA 22
The report opens but the charts are missing	The PNG was not saved in the same folder, or the picture was linked and not inserted.	Insert the saved PNG into the document and export to PDF again.	DA 22
The talk runs over five minutes	There are too many slides or too much code on screen.	Keep one idea per slide and cut the slide that explains the method in detail.	DA 23
The projector fails	The demonstration depends on a live notebook or the internet.	Use the PDF and the screenshots, and explain the workflow aloud.	DA 23
The notebook will not open on the day	The file or environment was not tested on the showcase laptop.	Use the PDF and screenshots, explain the workflow, and arrange a short live rerun afterwards.	DA 24
A question about the code cannot be answered	The student has not traced that line since they wrote it.	Say what the line is meant to do, then run it on a small input to show it.	DA 24